Se Yoon Lee, PhD

Lindenhurst, IL | (979) 888-4107 | seyoonlee.stat.math@gmail.com
Google Scholar | Website | LinkedIn | GitHub

Research Profile

Statistician combining a strong theoretical foundation in causal inference, semiparametric efficiency theory, and Bayesian analysis with professional experience at Big Pharma (AbbVie, Johnson & Johnson, and Amgen). Strong publication record in leading statistics/biostatistics journals and ML/AI conference proceedings.

Research areas: Causal inference; semiparametric efficiency; external controls and data borrowing; Bayesian analysis and theory; survival analysis; machine learning/artificial intelligence.

Education

PhD in Statistics, Texas A&M University May 2021
Advisor: Bani K. Mallick. Dissertation: *Bayesian Hierarchical Modeling: Application towards Complex and High-dimensional Data*.

MA in Applied Statistics, Yonsei University Feb 2016
Advisor: Joseph H. T. Kim. Thesis: *Exponentiated generalized Pareto distribution: an alternative to the generalized Pareto distribution.*

BS in Mathematics, Yonsei University Feb 2013
Graduated second in a class of 43 students.

Professional Experience (Permanent Positions)

AbbVie Feb 2025 – Present
Manager, Statistics *North Chicago, IL*

- Develop causal inference methods to augment clinical trials with real-world data and address external-control settings.
- Develop survival analysis methods for causal inference and real-world evidence applications.
- Lead and support observational studies and statistical work for health technology assessment (HTA) submissions within the real-world data group.

Edwards Lifesciences May 2023 – Feb 2025
Manager, Biostatistics *Irvine, CA*

- Developed R simulations and power calculations for Bayesian adaptive cardiovascular trial designs, and wrote statistical analysis plans for regulatory submissions.
- Implemented unblinded sample-size re-estimation using the promising-zone approach and CHW test statistics for Finkelstein–Schoenfeld hierarchical composite endpoints.
- Supported cross-functional teams in trial design, power analysis, and primary endpoint analyses; trained biostatisticians through seminars on Bayesian medical-device trials.

Johnson & Johnson Apr 2022 – May 2023
Senior Biostatistician *Irvine, CA*

- Developed Bayesian basket, power-prior, and group-sequential designs to evaluate medical-device safety and effectiveness, with a focus on cardiovascular disease.
- Led basket-trial design development and authored the statistical analysis plan for the SECURE post-market study (NCT04750798).
- Led machine-learning and statistical-methodology/consulting working groups; supported clinicians in power analysis, trial design, and primary endpoint analyses.

Amgen May 2021 – Apr 2022
Scientist – Modeling & Simulations *Thousand Oaks, CA*

- Developed pharmacokinetic, pharmacodynamic, and Cox regression models for Phase I AMG160 trials in metastatic castration-resistant prostate cancer.
- Investigated machine-learning and deep-neural-network models using single-dose data to simulate multiple-dosing scenarios.

Earlier Research & Teaching Experience

Novartis May – Aug 2020
Biostatistics Summer Intern *East Hanover, NJ*

- Developed a Bayesian linear mixed-effects model using HAWK and HARRIER data (approximately 1,800 patients) to predict visual acuity during maintenance treatment for wet age-related macular degeneration and inform personalized dosing.

EMD Serono, Merck KGaA May – Aug 2019
Pharmacometrics Summer Intern *Billerica, MA*

- Developed a Bayesian adaptive Phase I oncology trial design incorporating adverse-event toxicity grades to inform dose selection. [Poster]

Texas A&M University Aug 2016 – May 2021
Graduate Research/Teaching Assistant *College Station, TX*

- Developed and applied hierarchical, nonlinear mixed-effects, longitudinal, clustering, and classification models to biomedical, petroleum, wind-energy, and COVID-19 research.
- Served as a teaching assistant for undergraduate and graduate courses.

Peer-Reviewed Publications

*Corresponding author.

[1] **Se Yoon Lee*** (2026). Reliable Bayesian Causal Inference for Externally Controlled Single-Arm Trials Under Transportability Violations. *Statistics in Medicine*, 45(23–24), e70748.

[2] **Se Yoon Lee*** and Jae Kwang Kim (2026). MEC: Machine-Learning-Assisted Generalized Entropy Calibration for Semi-Supervised Mean Estimation. *Proceedings of the 43rd International Conference on Machine Learning*, PMLR 306, 64086–64135. [Conference] [OpenReview] [Video]

[3] **Se Yoon Lee*** (2026). Asymptotic validity of Schoenfeld's sample size formula for the Cox proportional hazards model via the Wald test approach. *Statistical Methods in Medical Research*, 35(4), 681–694.

[4] **Se Yoon Lee*** (2025). Power priors and type I error control: constrained borrowing of external control data. *Journal of Biopharmaceutical Statistics*.

[5] **Se Yoon Lee*** (2025). A note on the sample size formula for a win ratio endpoint. *Statistics in Medicine*, 44(15–17), e70165. [GitHub]

[6] **Se Yoon Lee*** (2025). Eliciting the discount parameter in a power prior method on the basis of the type I error consideration. *Statistics in Biopharmaceutical Research*, 17(2), 235–246.

[7] **Se Yoon Lee***, Peng Zhao, Debdeep Pati, and Bani K. Mallick (2024). Tail-adaptive Bayesian shrinkage. *Electronic Journal of Statistics*.

[8] **Se Yoon Lee*** (2024). Using Bayesian Statistics in Confirmatory Clinical Trials in the Regulatory Setting: A Tutorial Review. *BMC Medical Research Methodology*.

[9] **Se Yoon Lee*** (2023). A flexible dose-response modeling framework based on continuous toxicity outcomes in phase I cancer clinical trials. *Trials*.

[10] Abhinav Prakash, **Se Yoon Lee**, Xin Liu, Lei Liu, Bani Mallick, and Yu Ding* (2023). A Bayesian hierarchical model to understand the effect of terrain on wind turbine power curves. *IEEE Transactions on Sustainable Energy*.

[11] **Se Yoon Lee*** (2022). The Use of a Log-Normal Prior for the Student t-Distribution. *Axioms*.

[12] **Se Yoon Lee**[*] (2022). Bayesian Nonlinear Models for Repeated Measurement Data: An Overview, Implementation, and Applications. *Mathematics*.

[13] **Se Yoon Lee**, Alain Munafo, Pascal Girard, and Kosalaram Goteti[*] (2022). Optimization of dose selection using multiple surrogates of toxicity as a continuous variable in Phase I cancer trial. *Contemporary Clinical Trials*. [GitHub]

[14] **Se Yoon Lee**[*] (2021). Gibbs sampler and coordinate ascent variational inference: a set-theoretical review. *Communications in Statistics - Theory and Methods*.

[15] **Se Yoon Lee**[*] and Bani K. Mallick (2021). Bayesian Hierarchical modeling: application towards production results in the Eagle Ford Shale of South Texas. *Sankhyā: The Indian Journal of Statistics, Series B*. [GitHub]

[16] Kahkashan Afrin, Ashif Iquebal, Mostafa Karimi, Allyson Larsen, **Se Yoon Lee**, and Bani K. Mallick[*] (2020). Directionally Dependent Multi-View Clustering Using Copula Model. *PLOS ONE*. All authors contributed equally and are listed alphabetically by last name.

[17] **Se Yoon Lee**[*], Bowen Lei, and Bani K. Mallick (2020). Estimation of COVID-19 spread curves integrating global data and borrowing information. *PLOS ONE*. [GitHub] [R code]

[18] **Seyoon Lee** and Joseph H. T. Kim[*] (2018). Exponentiated generalized Pareto distribution: Properties and applications towards extreme value theory. *Communications in Statistics - Theory and Methods*, 48(8), 2014–2038.

Manuscripts with Revision or Resubmission Invited

[1] **Se Yoon Lee**[*] and Jae Kwang Kim (2026+). Statistical Optimality of Prediction-Powered Inference. Rejection with resubmission invited, *Electronic Journal of Statistics*. arXiv:2606.08730.

[2] **Se Yoon Lee**[*], Jieying Jiang, David Chen, Weili He, and Dai Feng (2026+). Applying Ensemble Survival Models to Predict Median Survival Time in Health Technology Assessment. Major revision invited, *Pharmaceutical Statistics*.

[3] **Se Yoon Lee**[*] (2026+). A Practical Roadmap for Establishing the Nonexistence of Uniformly Most Powerful Tests. Major revision invited, *Communications in Statistics – Simulation and Computation*.

Manuscripts Under Review

[1] **Se Yoon Lee**[*], Yonghyun Kwon, and Jae Kwang Kim (2026+). MEC-Cox: Machine-Learning-Assisted Generalized Entropy Calibration for ATT Marginal Hazard-Ratio Estimation. Under review. arXiv:2606.08305.

[2] **Se Yoon Lee**[*] (2026+). Understanding the Causal-Gap in Externally Controlled Single-Arm Trials: Study Design, Sensitivity Analysis, and Bayesian Borrowing. Under review.

[3] **Se Yoon Lee**[*] (2026+). Covariate adjustment in randomized trials: A review of semiparametric efficiency and inference. Under review, *Journal of the American Statistical Association*.

[4] **Se Yoon Lee**[*] (2026+). A Unified Semiparametric Framework for Prospective Causal-Gap Sensitivity Analysis Across Randomized, Externally Controlled, and Hybrid Trials. Under review, *Biometrics*.

[5] Yonghyun Kwon, **Se Yoon Lee**, and Jae Kwang Kim (2026+). Bregman Calibration for Transfer Learning under Covariate Shift. Under review.

[6] **Se Yoon Lee**[*], Jieying Jiang, Yulei He, Weili He, and Dai Feng (2026+). A Flexible Multiple Imputation Approach to Reduce Assessment Schedule Bias in Clinical Trial and Real-World Progression-Free Survival Data. Under review.

Editorial Service & Honors

Editorial Board Member, *BMC Medical Research Methodology*.

Peer review: NeurIPS; ICML (Gold Reviewer); ICLR; AISTATS; AAAI; *Artificial Intelligence Review*; *Nature Medicine*; *Statistics in Medicine*; *Biostatistics*; *Pharmaceutical Statistics*; *Statistical Methods in Medical Research*; *Bayesian Analysis*; *Contemporary Clinical Trials Communications*; *European Journal of Clinical Investigation*; *The Journal of Clinical Pharmacology*; *Frontiers in Public Health*; *The R Journal*; *Computational Geosciences*; *Journal of Applied Statistics*.

Springer Nature Editor of Distinction Awards May 2026
Recognized among the top 20% of editors for service to *BMC Medical Research Methodology*.

Second Place, Biopharmaceutical Section Poster Awards 2026
First-author poster: *A Flexible Multiple Imputation Approach to Reduce Assessment Schedule Bias in PFS Analysis.*

Anant Kshirsagar Fellowship (Jul 2018); **First Place, SETCASA Poster Session** (Apr 2018).

Programming & Languages

Programming and analytical tools: R, Python, SAS, SQL, NONMEM, Microsoft Access.

Languages: English and Korean (fluent); Chinese and Japanese (elementary).

References

Bani K. Mallick Distinguished Professor, Texas A&M University, bmallick@stat.tamu.edu

Debdeep Pati Professor, University of Wisconsin–Madison, dpati2@wisc.edu

Samiran Sinha Professor, Texas A&M University, sinha@stat.tamu.edu

Alan Dabney Professor, Texas A&M University, adabney@stat.tamu.edu